

Challenges (And How We Cope With Them?)

- Different IoT analytics → **resource consumption models**
 - Temperature average **versus** people-counting video analytics
- Heterogeneous devices → **analytics decomposition**
 - Raspberry Pi **versus** Xeon servers
- Time constraints → **polynomial-time algorithms**
 - Real-time requirements of trigger-action **or** IFTTT (if this, then that) rules

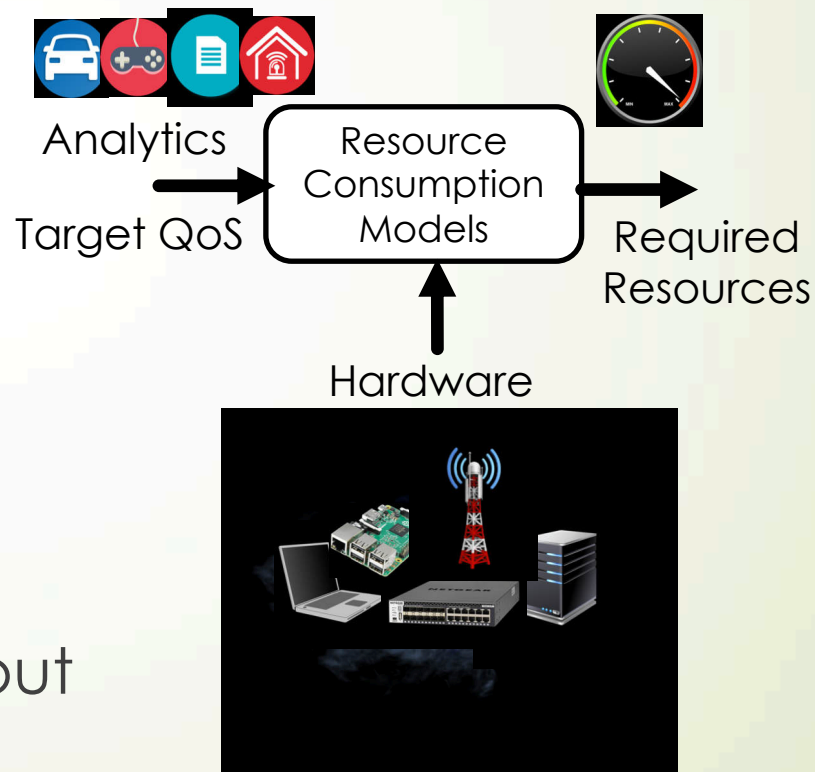
Resource Consumption Modeling

Input:

- IoT Analytics
- Target QoS
- Hardware Spec

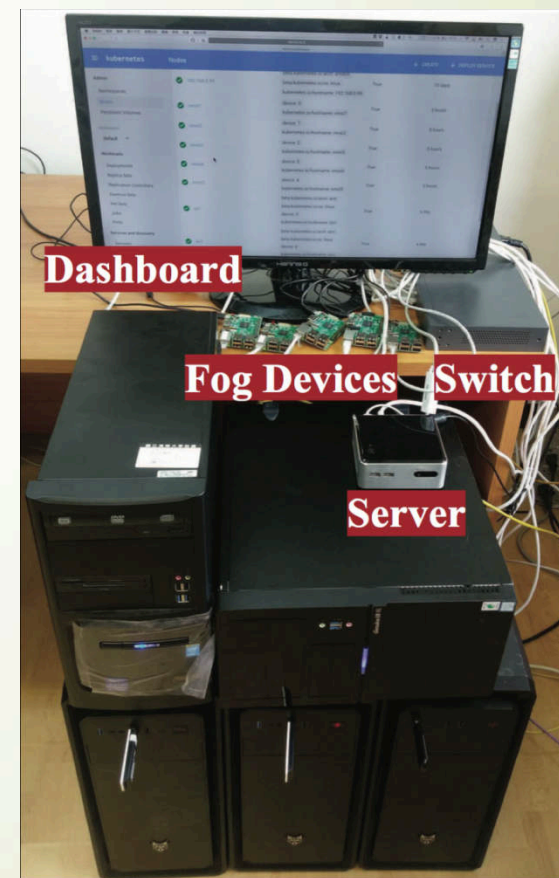
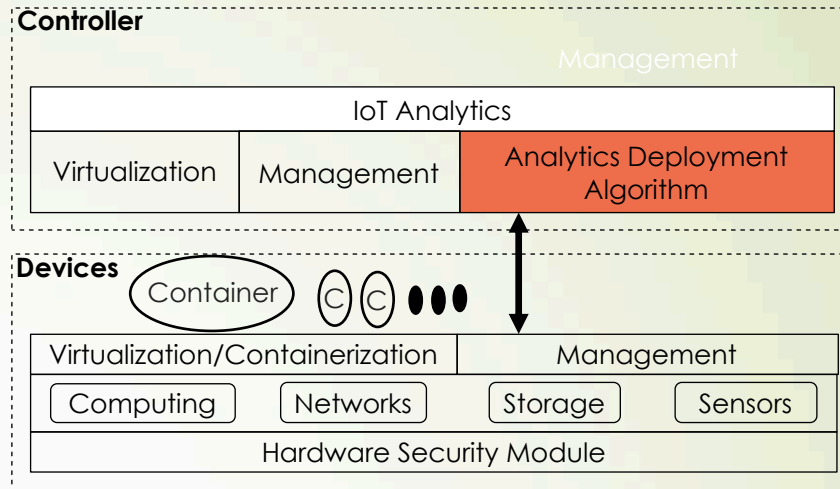
Output:

- CPU load
- RAM usage
- Network throughput



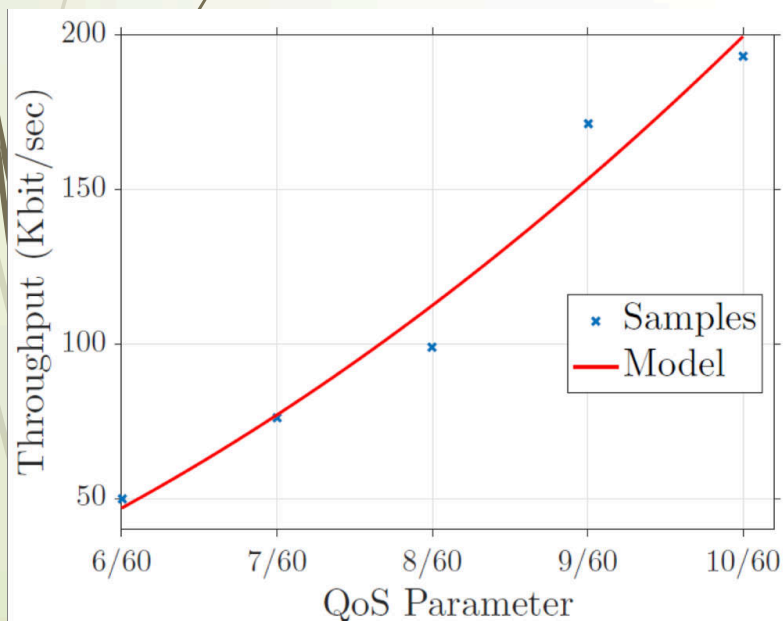
Testbed

- Devices: 5 Raspberry PI3s + 5 PCs (i7)
 - Docker + Kubernetes
- Server: a tiny PC (i5/8GB RAM)
 - Kubernetes
- Ethernet
- IoT Analytics (on TensorFlow)
 - Air quality monitor (QoS: 0.25 – 4 Hz)
 - Sound classifier (QoS: 6/60 – 10/60 Hz)
 - Object recognizer (QoS: 5/60 – 9/60 Hz)



Deriving the Models for Raspberry PIs

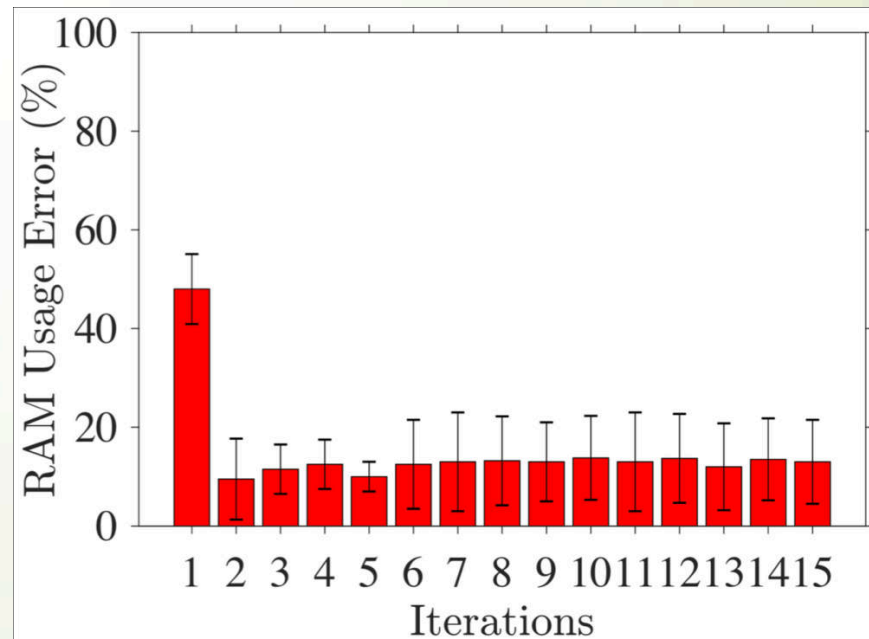
- QoS parameter
 - Frame rate (could be others)
- Power functions: CPU and network
- Constant: RAM



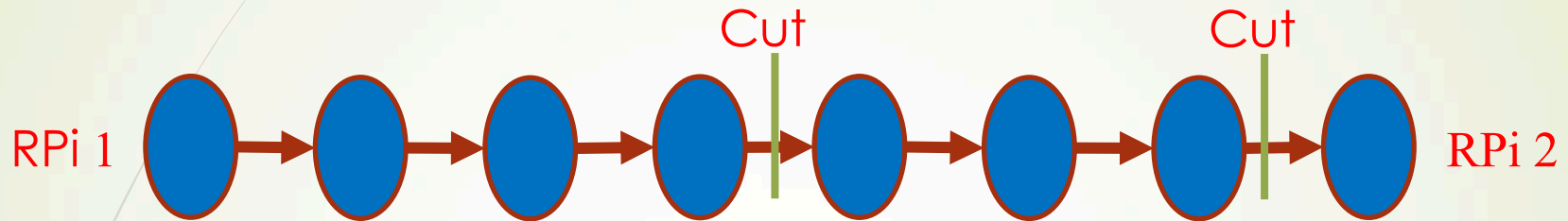
Analytics	CPU	Network
Air Quality Monitor	0.99	0.99
Sound Classifier	0.85	0.84
Object Recognizer	0.99	0.97

New Devices (Say a PC)?

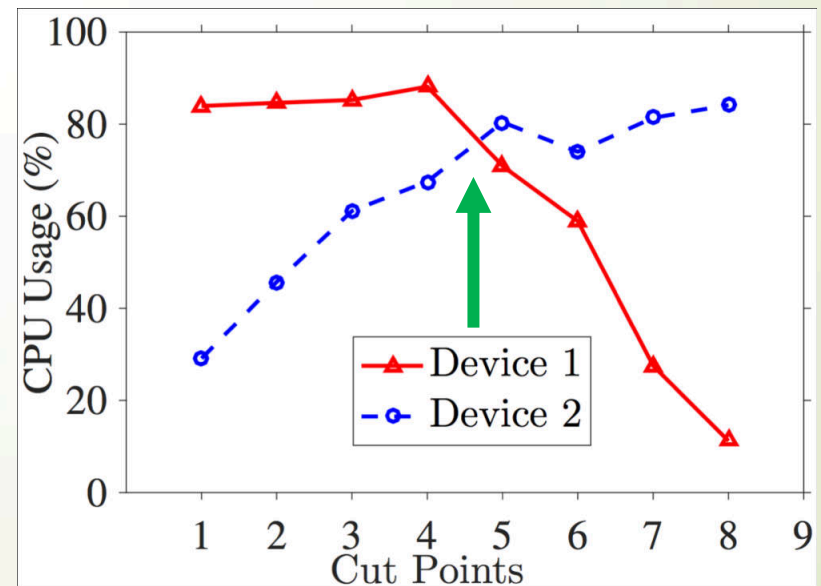
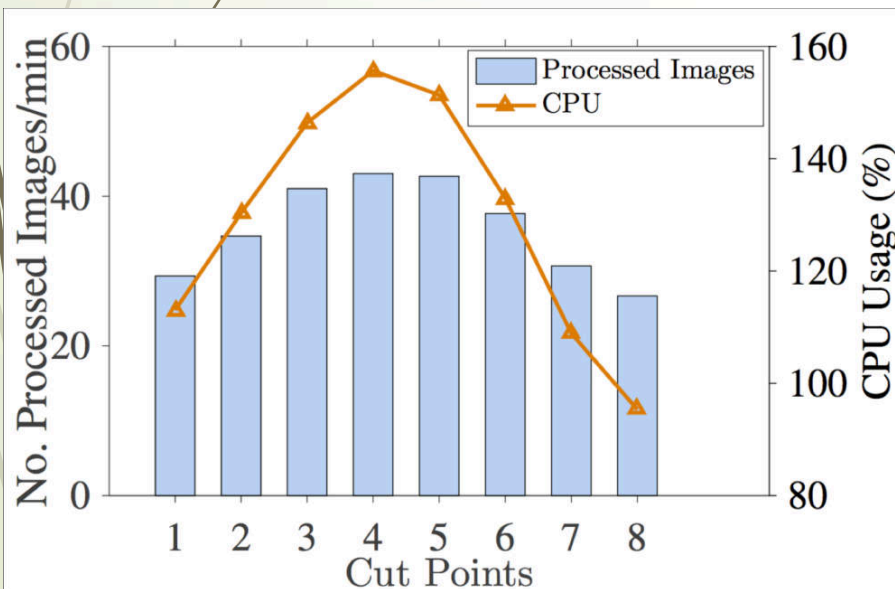
- Online regression
- Bootstrap from Raspberry Pis' system models
- 15 Iterations
- Mean error
 - CPU: 2%
 - RAM: 13%



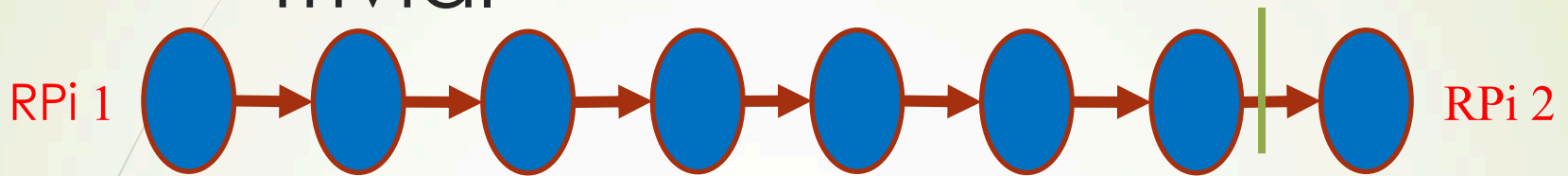
Analytics Decomposition is Possible



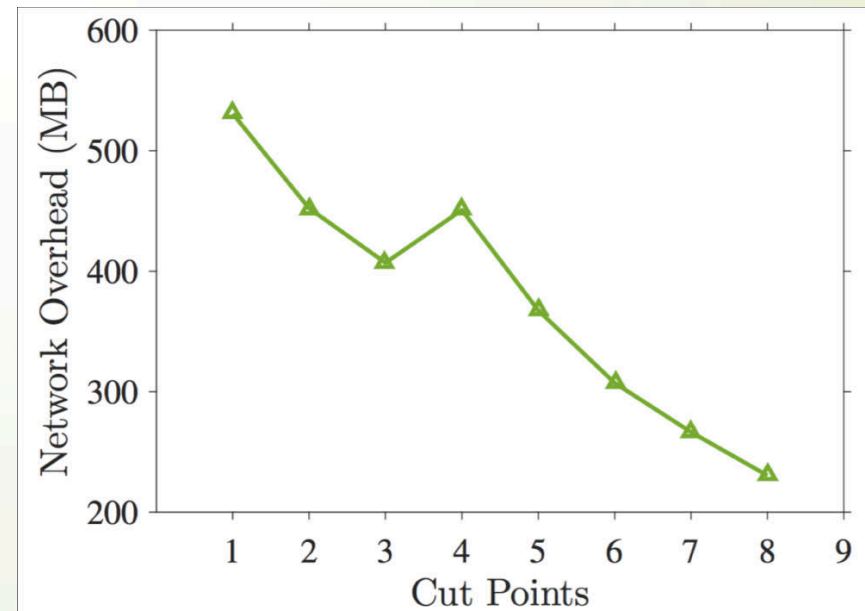
- Equally divide the complexity among devices maximizes the performance



But How to Cut Them is Non-Trivial

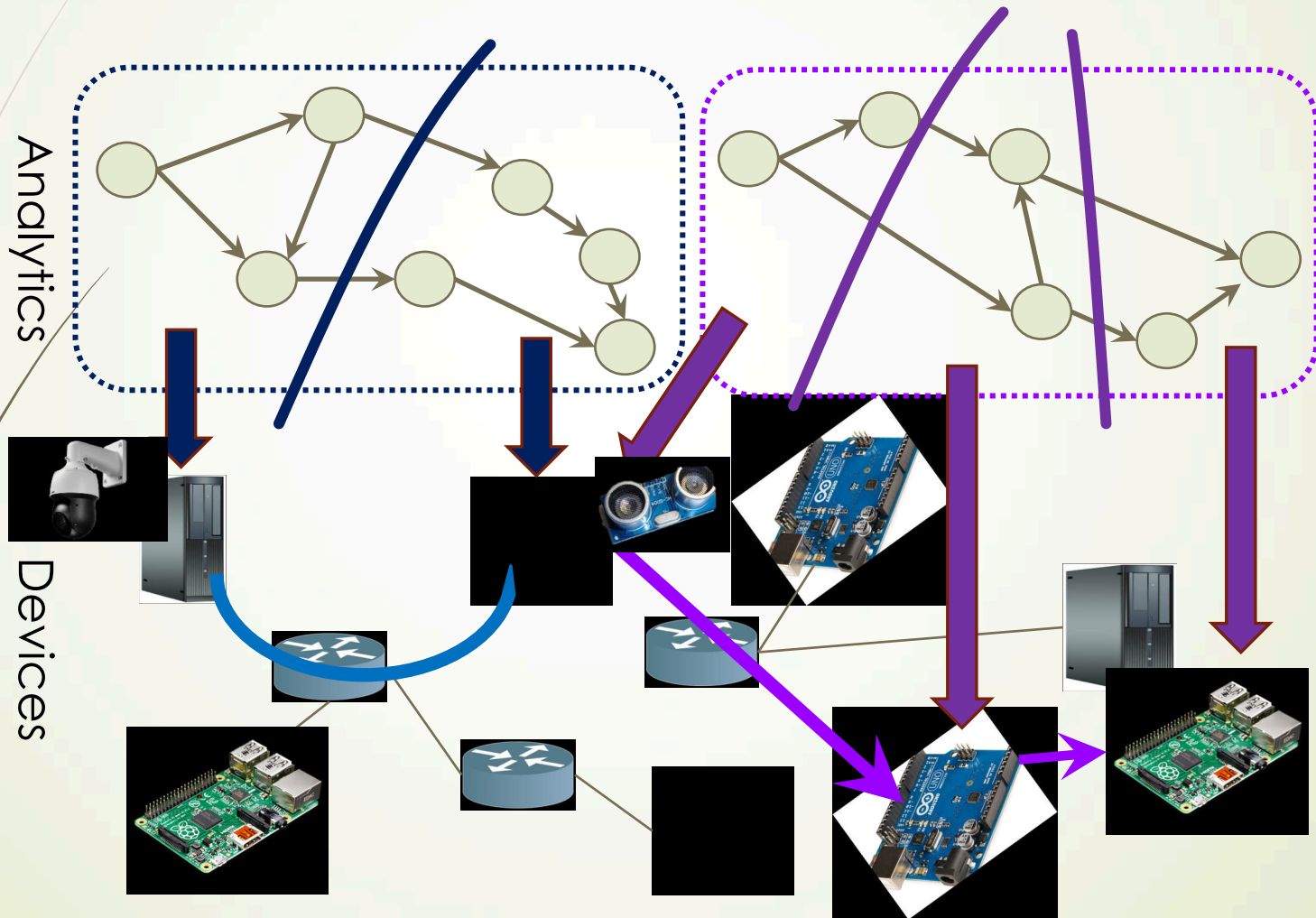


- Perform more computations on device 1
→ less network traffic
- Complex tradeoff between QoS versus resource
- The best tradeoff?



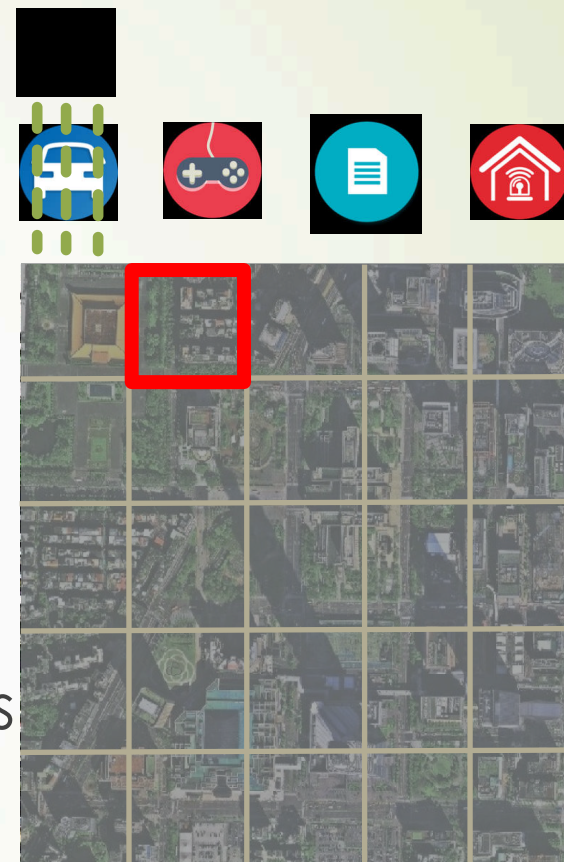
Back to The Analytics Deployment Problem

Goal: Maximize number of deployed analytics (requests)



Formulation #1 (More Tractable)

- Decomposable analytics
 - Arbitrary operator sizes
- Location-aware
 - One of the grids
- Reduced device heterogeneity
 - Proportional available resources
- Convert link constraints to node constraints



✓ NP hard

APproXimation Algorithm (APX)

1. Step 1: Analytics selection
 - Least total required resource first $\min_{q \in Q} \sum_{u \in U} F(q, u, \dots)$
 - q : request
 - u : resource type
 - F : model
2. Step 2: Device selection
 - Round robin
3. Step 3: Analytics decomposition
 - As many operators as possible

